SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	N.Thanveer basha	**Reg. No.:**	192372122
Section / Batch:	2023	**Date:**	22/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
		GRAND TOTAL:	**100 Marks**

SECTION A – APPLICATION BASED QUESTIONS

Q9. Application: Data Backup Optimization System

A backup system uses an algorithm with recurrence $T(n) = 2T(n/2) + \sqrt{n}$. Apply the Master Theorem to determine the time complexity. Compare f(n) with n^log_b a and identify the correct case. Analyze performance for large datasets. Interpret efficiency.

Q10. Application: Big Data Analytics Pipeline

A data analytics pipeline processes input using $T(n) = T(n-1) + \log n$. Apply the substitution method to solve the recurrence. Expand step-by-step and derive complexity. Analyze how the algorithm scales. Interpret efficiency using asymptotic notation.

Code of Conduct

I N. Thanveer Basha-192372122 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student N. Thanveer Basha

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Q9. Application: Data Backup Optimization System

A backup system uses an algorithm with recurrence T(n) = 2T(n/2) + √n. Apply the Master Theorem to determine the time complexity. Compare f(n) with n^log_b a and identify the correct case. Analyze performance for large datasets. Interpret efficiency.

Solution

Step 1: Write the Given Recurrence

$$T(n) = 2T\left(\frac{n}{2}\right) + \sqrt{n}$$

The standard form of the Master Theorem is

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

Step 2: Identify the Values

By comparing the given recurrence with the standard form,

- $a = 2$
- $b = 2$
- $f(n) = \sqrt{n} = n^{1/2}$

Step 3: Calculate $\boldsymbol{n^{\log_b a}}$

$$n^{\log_b a} = n^{\log_2 2}$$

Since,

$$\log_2 2 = 1$$

Therefore,

$$n^{\log_2 2} = n$$

Step 4: Compare $\boldsymbol{f(n)}$and $\boldsymbol{n^{\log_b a}}$

We have,

$$f(n) = n^{1/2}$$

and

$$n^{\log_b a} = n$$

Since

$$n^{1/2} = O(n^{1-\epsilon})$$

where

$$\epsilon = \frac{1}{2} > 0$$

the function $f(n)$is polynomially smaller than n.
Therefore, **Master Theorem Case 1** applies.

Step 5: Apply Master Theorem
According to **Master Theorem Case 1**,
If

$$f(n) = O(n^{\log_b a - \epsilon})$$

for some $\epsilon > 0$, then

$$T(n) = \Theta(n^{\log_b a})$$

Substituting the values,

$$T(n) = \Theta(n)$$

Time Complexity

$$\boxed{T(n) = \Theta(n)}$$

Performance Analysis

- The algorithm runs in **linear time**.
- The execution time increases proportionally with the amount of backup data.
- It efficiently handles large datasets without significant overhead.
- Therefore, it is suitable for large-scale data backup systems.

Interpretation
The Data Backup Optimization System has a **time complexity of** $\Theta(n)$. Since the running time grows linearly with the input size, the algorithm is efficient, scalable, and capable of processing large volumes of backup data with good performance.

Q10. Application: Big Data Analytics Pipeline

A data analytics pipeline processes input using T(n) = T(n-1) + log n. Apply the substitution method to solve the recurrence. Expand step-by-step and derive complexity. Analyze how the algorithm scales. Interpret efficiency using asymptotic notation.

Solution

Step 1: Write the Given Recurrence

$$T(n) = T(n-1) + \log n$$

Step 2: Expand the Recurrence

Substitute $T(n-1)$:

$$T(n) = [T(n-2) + \log(n-1)] + \log n$$
$$T(n) = T(n-2) + \log(n-1) + \log n$$

Again substitute $T(n-2)$:

$$T(n) = [T(n-3) + \log(n-2)] + \log(n-1) + \log n$$
$$T(n) = T(n-3) + \log(n-2) + \log(n-1) + \log n$$

Continuing this process,

$$T(n) = T(1) + \log 2 + \log 3 + \cdots + \log n$$

Step 3: Simplify the Summation

Using the logarithm property,

$$\log 2 + \log 3 + \cdots + \log n = \log(2 \times 3 \times \cdots \times n)$$
$$= \log(n!)$$

Using Stirling's Approximation,

$$\log(n!) = \Theta(n\log n)$$

Therefore,

$$T(n) = \Theta(n\log n)$$

Time Complexity

$$\boxed{T(n) = \Theta(n\log n)}$$

Performance Analysis

- The algorithm performs **logarithmic work** in each recursive step.
- There are $n$recursive calls.
- The overall running time is $\Theta(n\log n)$.
- The algorithm scales efficiently for large datasets compared to quadratic-time algorithms.

Interpretation

The Big Data Analytics Pipeline has a **time complexity of** $\Theta(n\log n)$. The running time increases slightly faster than linear time because each recursive step performs logarithmic work. This makes the algorithm efficient and suitable for processing large-scale datasets, providing good scalability and performance in real-world big data applications.